

MODULE <i>TCommit</i>	
CONSTANT <i>RM</i>	The set of participating resource managers
VARIABLE <i>rmState</i>	<i>rmState</i> [<i>rm</i>] is the state of resource manager <i>rm</i> .
$TCTypeOK \triangleq$ $rmState \in [RM \rightarrow \{\text{"working"}, \text{"prepared"}, \text{"committed"}, \text{"aborted"}\}]$	
$TCInit \triangleq rmState = [r \in RM \mapsto \text{"working"}]$	
$canCommit \triangleq \forall r \in RM : rmState[r] \in \{\text{"prepared"}, \text{"committed"}\}$	
$notCommitted \triangleq \forall r \in RM : rmState[r] \neq \text{"committed"}$	
$Prepare(r) \triangleq \wedge rmState[r] = \text{"working"}$ $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"prepared"}]$	
$Decide(r) \triangleq \vee \wedge rmState[r] = \text{"prepared"}$ $\wedge canCommit$ $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"committed"}]$ $\vee \wedge rmState[r] \in \{\text{"working"}, \text{"prepared"}\}$ $\wedge notCommitted$ $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"aborted"}]$	
$TCNext \triangleq \exists r \in RM : Prepare(r) \vee Decide(r)$	
$TCConsistent \triangleq$ $\forall r1, r2 \in RM : \neg \wedge rmState[r1] = \text{"aborted"}$ $\wedge rmState[r2] = \text{"committed"}$	
$TCSpec \triangleq TCInit \wedge \Box[TCNext]_{rmState}$	
THEOREM $TCSpec \Rightarrow \Box(TCTypeOK \wedge TCConsistent)$	